

Example 10.3

$$\text{ARIMA}(0, 0, 0) \times (0, 1, 1)_{12}$$

$$\text{Period} = 12$$

$$\text{Seasonal component: } P = 0, D = 1, Q = 1;$$

$$\text{Non-seasonal component: } p = 0, d = 0, q = 0.$$

Take lag 12 difference once.

No seasonal AR term

Seasonal MA(1) term

No non-seasonal terms

$$\phi_p(B) = 1; \quad \theta_q(B) = 1$$

$$\Phi_p(B^{12}) = 1; \quad \Theta_q(B^{12}) = 1 + \Theta_1 B^{12}$$

S-ARIMA model

$$\phi_p(B) \Phi_p(B^h) (1-B)^d (1-B^h)^D X_t$$

$$= \theta_q(B) \Theta_q(B^h) Z_t$$

$$\Rightarrow (1-B^{12}) X_t = (1 + \Theta_1 B^{12}) Z_t$$

$$\Rightarrow X_t - X_{t-12} = Z_t + \Theta_1 Z_{t-12}$$